

13.2 Gravitational Potential

Question Paper

Course	CIEA Level Physics
Section	13. Gravitational Fields
Topic	13.2 Gravitational Potential
Difficulty	Hard

Time allowed: 40

Score: /30

Percentage: /100

Question 1a

A spherical planet has a mass M and radius R . The planet may be considered to have all its mass concentrated at its centre.

A rocket, of mass m is launched from the surface of the planet.

The rocket engines are stopped when the rocket is at a height R above the surface of the planet, as shown in Fig 1.1.

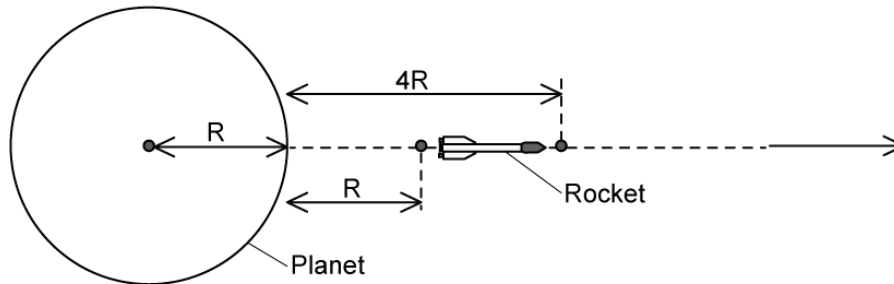


Fig. 1.1

The change in gravitational potential energy of the rocket as it travels from a height R to a height $4R$ above the planet's surface is equal to

$$\Delta E_p = x \left(\frac{GMm}{R} \right)$$

Determine the value of x

[3 marks]

Question 1b

At a distance R away from the surface of the planet the rocket has a velocity $2v$ and at a distance of $4R$ away from the planet a velocity v .

Show that the expression for the thermal energy lost ΔQ to the atmosphere of the planet as the rocket moves from R to $4R$ is given by

$$\Delta Q = \frac{3m}{2} \left(v^2 - \frac{GM}{5R} \right)$$

[4 marks]

Question 1c

The rocket travels towards a moon orbiting the planet

The distance from the centre of the planet to the centre of the moon is $200R$. The moon has a mass $\frac{M}{16}$ where M is the mass of the planet.

Determine the distance, in terms of R , from the centre of the planet at which

$$\frac{\Delta E_p}{\Delta r} = 0$$

Give your answer to the nearest whole number.

[4 marks]

Question 1d

On the axes of Fig. 1.2, sketch a graph to show the variation of the gravitational potential with distance along a line between the surface of the planet and the surface of the moon.

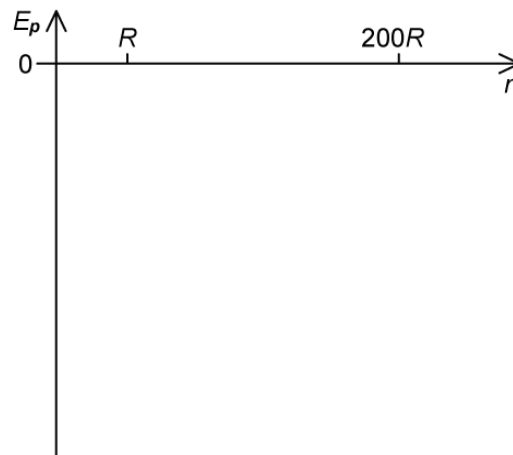


Fig. 1.2

[3 marks]

Question 2a

The Earth may be assumed to be a uniform sphere of radius $R = 6.38 \times 10^6 \text{ m}$, with its mass M concentrated at its centre.

A satellite of mass m orbits the Earth at a height h_1 above the Earth's surface.

Derive an expression for the total energy of the satellite in terms of G , M , m , h_1 and R .

[4 marks]

Question 2b

The satellite moves to a distance h_2 away from the surface of the Earth.

Derive an expression for the change in gravitational potential energy as the satellite moves from h_1 to h_2 .

[2 marks]**Question 2c**

When the satellite reaches a height h_2 above the surface of the Earth, it begins to drift out of its orbit.

Determine the distance moved by the satellite once it has drifted to the point the gravitational potential energy has increased by 20% of its initial value at h_1 .

Assume that the satellite moves along the same line connecting points h_1 , h_2 and the centre of the Earth.

Mass of the Earth = 5.97×10^{24} kg

Mass of the satellite = 1720 kg

Height $h_1 = 2.23 \times 10^6$ m

Height $h_2 = 3.05 \times 10^6$ m

[4 marks]

Question 3a

A space shuttle of mass $3 \times 10^6 \text{ kg}$ is travelling from the moon back to Earth. It accelerates uniformly from launch at 2.3 m s^{-2} . It has enough propellant to provide thrust for the first 155 seconds.

The mass of the moon is $7.35 \times 10^{22} \text{ kg}$ and the mean radius is 1740 km.

Calculate the work done by the rocket during the first 155 seconds after launch. State any assumptions you have made.

[3 marks]**Question 3b**

The Moon has a radius of approximately 27% that of the Earth, and a mass of 1.2% that of the Earth.

Show that the gravitational potential at the surface of the Earth is about 24 times greater than the gravitational potential at the surface of the moon.

[3 marks]



Model Answers: Hard

1a

(a) Derive an expression for the change in gravitational potential energy:

List the known quantities:

- Initial distance = $R + R = 2R$
- Final distance = $R + 4R = 5R$
- Mass of planet = M
- Mass of rocket = m

Write expressions for the initial and final gravitational potential energy:

- $E_p = -\frac{GMm}{r}$

- $\text{Initial GPE} = -\frac{GMm}{2R}$

AND

- $\text{Final GPE} = -\frac{GMm}{5R}$ [1 mark]

Consider the change in gravitational potential energy of the rocket:

- $\Delta E_p = \text{Final GPE} - \text{Initial GPE}$

- $\Delta E_p = \left(-\frac{GMm}{5R}\right) - \left(-\frac{GMm}{2R}\right)$ [1 mark]

- $\Delta E_p = \left(\frac{GMm}{2R}\right) - \left(\frac{GMm}{5R}\right) \Rightarrow \left(\frac{5GMm}{10R}\right) - \left(\frac{2GMm}{10R}\right)$

- $\Delta E_p = \frac{3GMm}{10R}$

- Hence $x = \frac{3}{10}$ [1 mark]

[Total: 3 marks]

1b

(b) Derive the expression for the thermal energy lost to the atmosphere of the planet:

List the known quantities:

- Initial velocity of rocket = $2v$
- Final velocity of rocket = v

Obtain an expression for the change in kinetic energy of the rocket:

- $\Delta KE = \text{Final KE} - \text{Initial KE}$
- $\Delta KE = \frac{1}{2}mv^2 - \frac{1}{2}m(2v)^2 = mv^2\left(\frac{1}{2} - \frac{4}{2}\right)$
- $\Delta KE = -\frac{3}{2}mv^2$ [1 mark]

Consider the conservation of energy:

- $\Delta E_p + \Delta KE + \Delta Q = 0$ [1 mark]

Substitute the known expressions and rearrange to make ΔQ the subject:

- Change in GPE: $\Delta E_p = \frac{3GMm}{10R}$ (from (a))
- $\Delta Q = -\Delta E_p - \Delta KE$
- $\Delta Q = -\frac{3GMm}{10R} - \left(-\frac{3mv^2}{2}\right)$ [1 mark]

- $\Delta Q = \frac{3mv^2}{2} - \frac{3GMm}{10R}$
- $\Delta Q = \frac{3m}{2} \left(v^2 - \frac{GM}{5R} \right)$ [1 mark]

[Total: 4 marks]

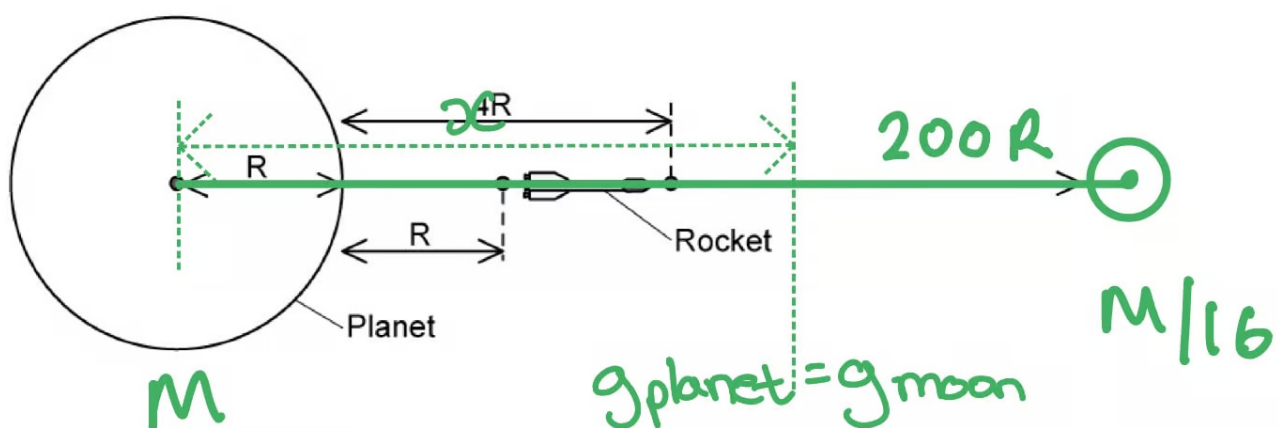
1c

(c) Calculate the distance from the planet's surface where the resultant gravitational field strength is zero:

List the known quantities:

- Mass of planet = M
- Mass of moon = $\frac{M}{16}$
- Radius of planet = R
- Distance from centre of planet to centre of moon = $200R$
- Distance from the surface of the planet to point where resultant gravitational field strength is zero = x

Write expressions for the gravitational field strength of the rocket and the moon:



- Gravitational field strength is equal to the gradient of the potential-distance graph i.e. $g = \frac{\Delta E_p}{\Delta r}$

- Hence, the point where $\frac{\Delta E_p}{\Delta r} = 0$ is where $g_{planet} = -g_{moon}$
- Field strength due to planet: $g_p = \frac{GM}{x^2}$ [1 mark]
- Field strength due to moon: $g_M = \frac{G\left(\frac{M}{16}\right)}{(200R - x)^2}$ [1 mark]

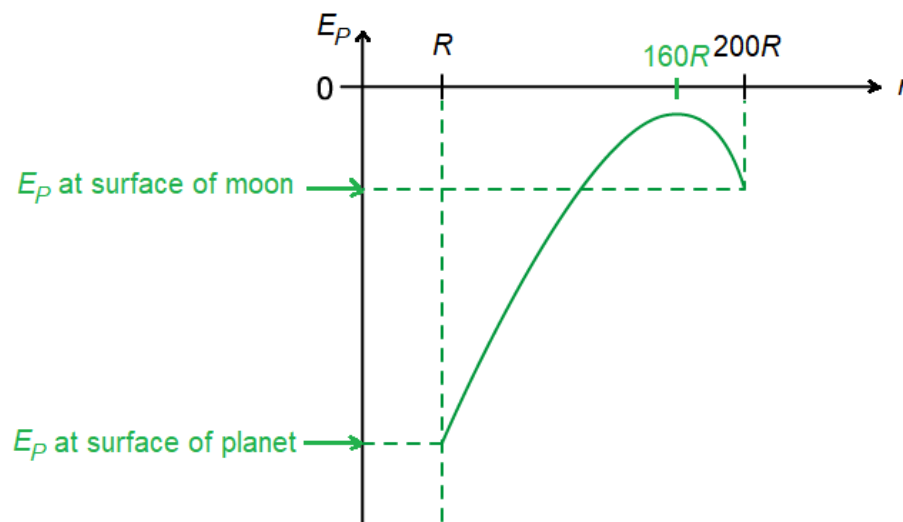
Equate the expressions for gravitational field strength to obtain a value for x :

- $g_p = -g_M$
- $\frac{GM}{x^2} = -\frac{GM}{16(200R - x)^2}$ [1 mark]
- $x^2 = 16(200R - x)^2 \Rightarrow \sqrt{x^2} = \sqrt{16(200R - x)^2}$
- $x = 4(200R - x) = 800R - 4x$
- $x = \frac{800}{5}R = 160R$ [1 mark]

[Total: 4 marks]

1d

(d) The variation of gravitational potential with distance between the planet and moon is as follows:



- Curve drawn starting at R (surface of planet) at maximum (negative) value of E_P ; [1 mark]
- Peak at $160R$ at minimum (negative) value of E_P close to (but below) zero; [1 mark]
- Curve drawn finishing at $200R$ (surface of moon) slightly below peak; [1 mark]

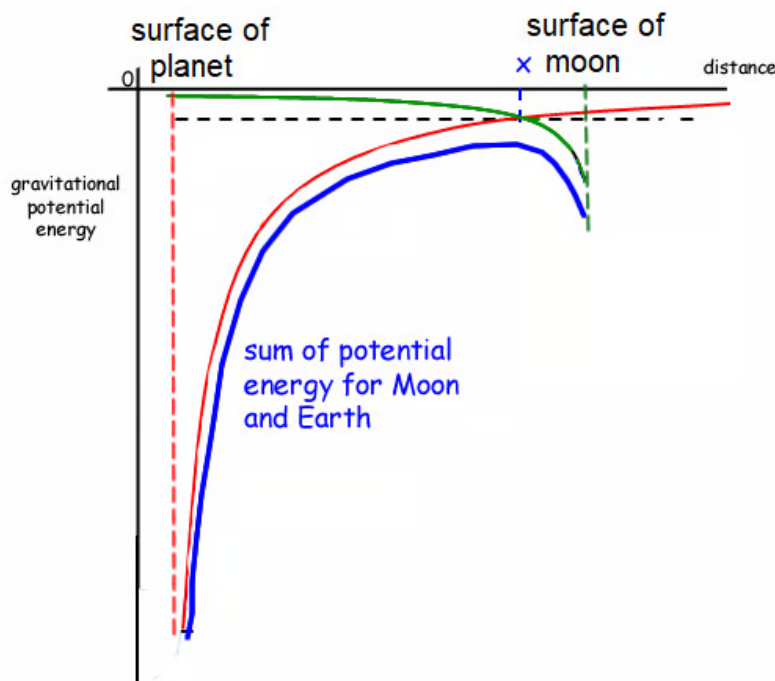
[Total: 3 marks]

Since both the moon and the planet create a gravitational field, points between them have a gravitational potential influenced by both bodies. The gravitational potential increases away from a body (the planet), however, decreases again closer to another body which creates a gravitational field, which is the moon. Therefore, the graph you need to sketch is a combination of the gravitational potential curves for the planet and the moon.

In the graph you have to sketch, the (negative) value of the combined curve at any point measures the amount of energy you would need to get away from that place in space to infinity where you would be free of the influence of both masses. Each point will therefore be the sum of the two curves:

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In the graph you have to sketch, the (negative) value of the combined curve at any point measures the amount of energy you would need to get away from that place in space to infinity where you would be free of the influence of both masses. Each point will therefore be the sum of the two curves:



The peak of your graph will be at the place, somewhere between the two, where you require the least amount of energy to escape the planet-moon system. This is the point where

$$g_{moon} = -g_{planet}$$

Remember the 'potential gradient' is equal to

$$g = \frac{\Delta E_p}{\Delta r}$$

Therefore, the two values of g will cancel when the gradient of the gravitational

potential graph is zero – this is at the distance you calculated in part (c).

Make sure you understand that the gravitational potential energy doesn't cancel out at this point of zero force as you will always need energy to escape to infinity from the combined force of attraction from the two masses. That amount of energy is the difference between the top of the curve and the axis, therefore if you draw your curve to zero you will miss out on a mark, if you have a mean examiner you might even be given a 0 for the question!

2a

(a) Derive an expression for its total energy in terms of G , M , m , h_1 and R :

- Radius of the satellite's orbit: $r = (R + h_1)$

Obtain an expression for the linear speed v of the satellite:

- Centripetal force = gravitational force
- $F = \frac{mv^2}{r} = \frac{GMm}{r^2}$
- Speed of the satellite: $v = \sqrt{\frac{GM}{r}}$ [1 mark]

Write an expression for kinetic energy:

- Kinetic energy: $KE = \frac{1}{2}mv^2$
- $KE = \frac{1}{2}m \left(\sqrt{\frac{GM}{r}} \right)^2 = \frac{1}{2}m \left(\frac{GM}{r} \right)$
- $KE = \frac{GMm}{2r}$ [1 mark]

Write an expression for the kinetic energy + gravitational potential energy:

- Gravitational potential energy: $E_p = -\frac{GMm}{r}$
- Total energy: $E_T = KE + E_p$
- $E_T = \left(\frac{GMm}{2r}\right) + \left(-\frac{GMm}{r}\right)$ [1 mark]
- $E_T = -\frac{GMm}{2r} = -\frac{GMm}{2(R + h_1)}$ [1 mark]

[Total: 4 marks]

Even though the final expression must be in terms of h_1 and R , it's much easier to keep the distance as r until the very end to avoid errors!

2b

(b) Write an expression for the change in gravitational potential energy:

- Initial orbital radius: $r_1 = (R + h_1)$
- Final orbital radius: $r_2 = (R + h_2)$
- Initial $E_p = -\frac{GMm}{r_1} = -\frac{GMm}{(R + h_1)}$
- Final $E_p = -\frac{GMm}{r_2} = -\frac{GMm}{(R + h_2)}$

Obtain an expression for the change in gravitational potential energy between the two points:

- $\Delta E_p = \text{Final } E_p - \text{Initial } E_p$ [1 mark]
- $\Delta E_p = \left(-\frac{GMm}{r_2}\right) - \left(-\frac{GMm}{r_1}\right) = \frac{GMm}{r_1} - \frac{GMm}{r_2}$

- $\Delta E_p = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$
- $\Delta E_p = GMm \left(\frac{1}{R + h_1} - \frac{1}{R + h_2} \right)$ [1 mark]

[Total: 2 marks]

2c

(c) Calculate how far the satellite will have drifted at the point where its gravitational potential energy has decreased by 20%:

List the known quantities:

- Radius of Earth, $R = 6.38 \times 10^6 \text{ m}$
- Height, $h_1 = 2.23 \times 10^6 \text{ m}$
- Height, $h_2 = 3.05 \times 10^6 \text{ m}$
- Distance drifted out of orbit = x

Obtain an expression linking gravitational potential energy and orbital radius:

- G, M and m remain constant, hence: $E_p = -\frac{GMm}{r} \Rightarrow E_p \propto -\frac{1}{r}$
- Initial distance from centre of Earth:
 $r_i = (R + h_1) = (6.38 + 2.23) \times 10^6 = 8.61 \times 10^6 \text{ m}$
- Final distance from centre of Earth:
 $r_f = x + (R + h_2) = x + (6.38 + 3.05) \times 10^6 = x + (9.43 \times 10^6) \text{ m}$
- Initial GPE: $E_{p(i)} \propto \frac{1}{r_i}$ **AND** final GPE: $E_{p(f)} \propto \frac{1}{r_f}$ [1 mark]

Obtain an expression linking the orbital radius before and after drifting:

- When the satellite moves from r_i to r_f , the GPE **increases** by 20%:

$$E_{p(f)} = 1.2E_{p(i)}$$

2c

(c) Calculate how far the satellite will have drifted at the point where its gravitational potential energy has decreased by 20%:

List the known quantities:

- Radius of Earth, $R = 6.38 \times 10^6 \text{ m}$
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 $r_f = x + (R + h_2) = x + (6.38 + 3.05) \times 10^6 = x + (9.43 \times 10^6) \text{ m}$
- Initial GPE: $E_{p(i)} \propto \frac{1}{r_i}$ **AND** final GPE: $E_{p(f)} \propto \frac{1}{r_f}$ [1 mark]

Obtain an expression linking the orbital radius before and after drifting:

- When the satellite moves from r_i to r_f , the GPE **increases** by 20%:
 $E_{p(f)} = 1.2E_{p(i)}$
- Hence, $1.2E_{p(i)} = E_{p(f)} \propto \frac{1}{1.2r_i} = \frac{1}{r_f}$
- Therefore, $1.2r_i = r_f \Rightarrow r_i = 0.8r_f$ [1 mark]

Calculate the drift distance of the satellite, x :

- $8.61 \times 10^6 = 0.8(x + 9.43 \times 10^6)$

- $x = \frac{8.61 \times 10^6}{0.8} - (9.43 \times 10^6)$ [1 mark]
- $x = 1.33 \times 10^6 \text{ m}$ [1 mark]

[Total: 4 marks]

It takes longer but will get you the same answer if you:

- First, calculate the **initial GPE**

$$E_{p(i)} = - \frac{(6.67 \times 10^{-11})(5.97 \times 10^{24})(1720)}{(6.38 + 2.23) \times 10^6}$$

$$E_{p(i)} = -7.955 \times 10^{10} \text{ J}$$

- Then, calculate the **change in GPE** (you can ignore the minus signs!)

$$\Delta E_p = |E_{p(f)} - E_{p(i)}|$$

$$\Delta E_p = 1.2 \times (7.955 \times 10^{10}) - (7.955 \times 10^{10})$$

$$\Delta E_p = 1.591 \times 10^{10} \text{ J}$$

- Finally, **rearrange** the change in GPE equation to calculate x

$$\Delta E_p = GMm \left(\frac{1}{r_i} - \frac{1}{r_f} \right)$$

$$\frac{\Delta E_p}{GMm} = \frac{1}{r_i} - \frac{1}{r_f} \Rightarrow \frac{1}{r_f} = \frac{1}{r_i} - \frac{\Delta E_p}{GMm}$$

$$\frac{1}{r_f} = \frac{1}{r_i} - \frac{(1.591 \times 10^{10})}{(6.67 \times 10^{-11})(5.97 \times 10^{24})(1720)}$$

$$\frac{1}{r_f} = \frac{1}{(6.38 + 2.23) \times 10^6} - \frac{(1.591 \times 10^{10})}{(6.67 \times 10^{-11})(5.97 \times 10^{24})(1720)}$$

$$\frac{1}{r_f} = 9.291 \times 10^{-8} \Rightarrow r_f = 1.076 \times 10^7 \text{ m}$$

$$r_f = (1.076 \times 10^7) = x + (6.38 + 3.05) \times 10^6$$

$$x = (1.076 \times 10^7) - (6.38 + 3.05) \times 10^6$$

$$x = 1.33 \times 10^6 \text{ m}$$

3a

(a) Calculate the work done by the rocket during the first 155 seconds after launch:

List the known quantities:

- Mass of space shuttle, $m_s = 3.00 \times 10^6 \text{ kg}$
- Mass of the moon, $M_M = 7.35 \times 10^{22} \text{ kg}$
- Mean radius of the moon, $r_m = 1740 \text{ km} = 1.74 \times 10^6 \text{ m}$
- Gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
- Acceleration of space shuttle, $a = 2.3 \text{ m s}^{-2}$
- Time thrust is applied for, $t = 155 \text{ s}$
- Initial velocity of the space shuttle, $u = 0 \text{ m s}^{-1}$

Calculate the height of the space shuttle after 155 s:

- Uniform acceleration: $s = ut + \frac{1}{2}at^2$
- $s = 0 + \frac{2.3 \times 155^2}{2}$
- $s = 27\,629 = 2.763 \times 10^4 \text{ m}$ [1 mark]

Write expressions for the initial and final gravitational potential:

- Gravitational potential: $\phi = -\frac{GM}{r}$
- $\phi_{\text{initial}} = -\frac{GM_M}{r_m}$
- $\phi_{\text{final}} = -\frac{GM_M}{(s + r_m)}$

Calculate the gravitational potential difference, $\Delta\phi$, in the first 155 s:

- $\Delta\phi = \phi_{final} - \phi_{initial}$
- $\Delta\phi = \left(-\frac{GM_M}{s + r_m} \right) - \left(-\frac{GM_M}{r_m} \right)$
- $\Delta\phi = GM_M \left(\frac{1}{r_m} - \frac{1}{(s + r_m)} \right)$
- $\Delta\phi = (6.67 \times 10^{-11}) \times (7.35 \times 10^{22}) \times \left(\frac{1}{2.763 \times 10^4} - \frac{1}{(2.763 \times 10^4) + (1.74 \times 10^6)} \right)$
- $\Delta\phi = 1.75 \times 10^8 \text{ J kg}^{-1}$ [1 mark]

Calculate the work done by the space shuttle in the first 155 s:

- $E_p = \phi \times m \Rightarrow W = \Delta\phi \times m_s$
- $W = (1.75 \times 10^8) \times (3 \times 10^6)$
- $W = 5.25 \times 10^{14} \text{ J}$ [1 mark]

[Total: 3 marks]

3b

(b) Calculate the gravitational potential at the surface of the Earth:

List the known quantities

- Radius of the Earth, $r_E = 0.27 r_m$
- Mass of the Earth, $M_E = 0.012 M_m$
- Mass of the moon, $M_M = 7.35 \times 10^{22} \text{ kg}$
- Mean radius of the moon, $r_m = 1740 \text{ km} = 1.74 \times 10^6 \text{ m}$ (from **(a)**)
- Gravitational constant, $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ (from **(a)**)

Determine an expression for the gravitational potential on the Earth's surface:

- $r_M = 0.27r_E \Rightarrow r_E = 3.7r_M$
- $M_M = 0.012M_E \Rightarrow M_E = 83.3M_M$
- Gravitational potential: $\phi = -\frac{GM}{r}$
- $\phi_E = -\frac{GM_E}{r_E} = -\frac{G \times (83.3M_M)}{3.7r_M}$ [1 mark]

Write ϕ_E in terms of the gravitational potential on the surface of the moon:

- Gravitational potential on the moon's surface, $\phi_M = -\frac{GM_M}{r_M}$
- Divide ϕ_E by ϕ_M : $\frac{\phi_E}{\phi_M} = \left(-\frac{G(83.3M_M)}{3.7r_M} \right) \times \left(-\frac{r_M}{GM_M} \right)$
- Simplify: $\frac{\phi_E}{\phi_M} = \frac{83.3}{3.7}$ [1 mark]
- Therefore $\phi_E = 22.51\phi_M \approx 24\phi_M$ [1 mark]

[Total: 3 marks]

Read the question **carefully**! You have been given the information the **wrong way around**. You need to express the surface of the Earth in terms of the surface of the moon. So **swap** your values for radius and mass around!

You can also complete this question by calculating the values of ϕ_E and ϕ_M if you know the mass and radius of the Earth

Working this question out **algebraically** is **less prone** to calculator error.